

LLOYDMINSTER, AB, Canada

WMO: 718710

Lat: 53.3120N

Lon: 110.0744W

Elev: 2194

StdP: 13.57

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.5	-20.5	-34.1	0.9	-25.3	-28.7	1.2	-20.3	27.8	22.8	23.6	21.5	8.7	280	0.728

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	82.5	63.4	79.4	62.2	76.3	60.8	66.8	78.1	64.8	75.3	62.9	72.8	10.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	92.4	71.3	60.8	86.3	69.9	58.9	80.7	67.6	32.7	78.4	31.0	75.8	29.6	72.9	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	20.9	18.7	-30.9	88.7	5.8	4.2	-35.1	91.8	-38.5	94.2	-41.8	96.6	-46.0	99.6
			-31.2	70.9	5.7	2.5	-35.3	72.7	-38.7	74.1	-41.9	75.5	-46.1	77.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.9	6.9	9.7	20.6	37.5	50.4	58.4	62.9	60.9	51.9	38.2	21.6	9.8		
	DBStd	23.05	16.14	13.64	13.69	10.05	8.26	5.88	5.29	6.12	8.17	8.85	12.12	14.47		
	HDD50	6419	1335	1128	912	388	97	8	0	3	74	375	852	1247		
	HDD65	10713	1800	1548	1377	826	454	210	102	155	397	830	1302	1712		
	CDD50	1262	0	0	0	11	111	259	401	340	130	10	0	0		
	CDD65	82	0	0	0	0	3	11	38	27	3	0	0	0		
	CDH74	1194	0	0	0	4	94	176	428	380	111	1	0	0		
	CDH80	242	0	0	0	0	14	31	87	88	22	0	0	0		
	WSAvg	9.8	9.9	9.8	10.3	10.6	10.4	9.4	8.9	8.6	9.6	10.1	10.0	9.9		
	PrecAvg	16.1	0.7	0.5	0.6	1.1	1.8	2.9	3.1	2.3	1.3	0.7	0.6	0.7		
Precipitation	PrecMax	23.3	1.5	1.2	1.6	3.3	4.8	5.9	9.2	6.7	3.7	1.7	1.9	1.3		
	PrecMin	8.7	0.2	0.1	0.2	0.1	0.2	0.5	0.6	0.2	0.1	0.1	0.0	0.2		
	PrecStd	3.5	0.4	0.3	0.3	0.9	1.2	1.2	1.8	1.2	1.0	0.4	0.5	0.3		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	39.8	50.9	71.8	81.7	83.5	87.0	88.0	83.2	69.3	51.5	42.3		
		MCWB	35.1	34.7	41.4	51.8	57.6	63.0	67.0	65.4	60.9	53.9	42.9	36.0		
	2%	DB	37.2	35.6	44.1	65.1	76.3	78.8	81.9	81.8	76.9	61.3	44.6	37.7		
		MCWB	33.5	32.0	37.3	48.6	55.4	60.8	65.8	63.4	58.3	49.2	38.5	33.5		
	5%	DB	33.6	32.2	40.0	59.3	71.4	75.0	78.7	78.2	72.0	56.5	40.0	33.7		
		MCWB	30.8	29.3	35.3	45.1	53.2	59.1	64.2	62.4	56.3	46.0	35.4	30.6		
	10%	DB	29.0	28.7	37.0	53.9	67.0	71.5	75.4	74.4	66.9	51.8	36.2	29.3		
		MCWB	26.6	26.2	33.3	42.7	51.2	57.4	62.8	61.2	54.1	43.6	32.7	26.8		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.8	35.6	42.3	53.2	60.2	66.3	70.7	68.4	62.4	54.9	43.5	36.6		
		MCDB	39.5	39.0	49.3	69.4	75.8	77.3	81.3	80.8	77.7	67.9	50.2	41.2		
	2%	WB	33.6	32.2	38.0	49.3	57.0	63.2	67.7	65.8	59.6	50.3	39.3	33.7		
		MCDB	37.0	35.0	43.2	63.7	71.7	74.1	79.4	77.2	73.8	59.0	44.3	37.5		
	5%	WB	30.7	29.5	35.5	46.1	54.7	61.1	65.8	63.9	57.3	47.0	35.8	30.6		
		MCDB	33.7	32.2	39.5	58.0	68.7	71.1	76.3	75.1	70.4	55.0	39.4	33.7		
	10%	WB	26.6	26.3	33.5	43.1	52.4	59.2	63.9	62.0	54.9	44.2	33.0	26.9		
		MCDB	28.9	28.8	36.8	52.9	64.9	68.4	73.0	72.4	65.4	51.2	36.0	29.1		
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.9	15.5	19.5	22.5	19.9	20.4	21.2	21.4	17.9	13.7	14.0		
		MCDBR	16.8	18.8	17.7	28.8	29.1	26.1	25.5	28.0	30.4	26.1	18.1	17.9		
		MCWBR	14.1	16.4	12.7	16.0	13.9	12.4	12.6	13.0	14.9	15.6	13.0	15.0		
	5% WB	MCDBR	17.2	18.4	17.1	27.6	25.9	22.1	23.2	24.5	27.9	24.7	16.8	17.9		
		MCWBR	14.7	16.3	12.7	16.1	13.0	11.8	12.5	12.6	14.4	15.4	12.5	15.3		
Clear-Sky Solar Irradiance	taub		0.239	0.287	0.283	0.324	0.346	0.363	0.361	0.370	0.308	0.273	0.247	0.215		
	taud		2.338	2.206	2.382	2.382	2.368	2.346	2.378	2.332	2.489	2.555	2.453	2.366		
	Ebn at Noon		245	257	290	289	286	281	279	270	278	269	242	237		
	Edh at Noon		17	28	29	34	36	37	36	35	26	20	16	14		
All-Sky Solar Radiation	RadAvg		315	652	1129	1492	1843	1879	1948	1585	1119	650	311	232		
	RadStd		32	44	87	134	137	136	112	123	122	97	36	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+3.87	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page